

Normalization

Database Normalization

- **Normalization:** The process of structuring data to minimize duplication and inconsistencies.
- The process usually involves breaking down a **single Table** into two or more tables and defining relationships between those tables.
- Normalization is usually done in stages, with each stage applying some rules to the types of information which can be stored in a table.

Well-Structured Relations

- Goal is to avoid anomalies
 - **Insertion Anomaly** – adding new rows forces user to create duplicate data
 - **Deletion Anomaly** – deleting rows may cause a loss of data that would be needed for other future rows
 - **Modification Anomaly** – changing data in a row forces changes to other rows because of duplication

a table should not have more than one entity type

Example

<u>SID</u>	Sname	Bdate	City	ZipCode	<u>Subject</u>	Grade	Teacher
1	Ahmed	1/1/1980	Cairo	1010	DB	A	Hany
1	Ahmed	1/1/1980	Cairo	1010	Math	B	Eman
1	Ahmed	1/1/1980	Cairo	1010	WinXP	A	khalid
2	Ali	1/1/1983	Alex	1111	DB	B	Hany
2	Ali	1/1/1983	Alex	1111	SWE	B	Heba
3	Mohamed	1/1/1990	Mansoura	1210	NC	C	Mona

Question – What’s the primary key? Answer – Composite: SID, Subject

Why do these anomalies exist?

Because we’ve combined two themes (entity types) into one relation. This results in duplication, and an unnecessary dependency between the entities

Functional dependency

- a constraint between two attributes (columns) or two sets of columns
- $A \rightarrow B$ if “for every valid instance of A, that value of A uniquely determines the value of B”
- or ... $A \rightarrow B$ if “existing of B depending on a value of A”

... functional dependency

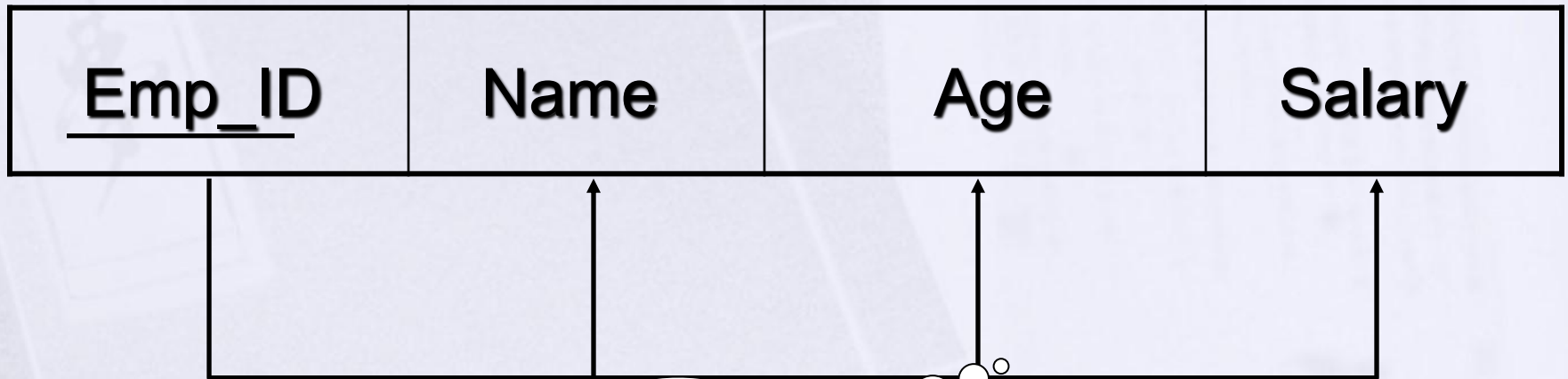
some examples

- social security number determines employee name
SSN $\rightarrow$ ENAME
- project number determines project name and location
PNUMBER $\rightarrow$ {PNAME, PLOCATION}
- employee ssn and project number determines the hours per week that the employee works on the project
{SSN, PNUMBER} $\rightarrow$ HOURS

keys and dependencies

EMPLOYEE1 (Emp_ID, Name, Age, Salary)

determinant



functional
dependency

Types of functional dependency

- **Full Functional Dependency**

Attribute is fully Functional Dependency on a PK if its value is determined by the **whole PK**

- **Partial Functional Dependency**

Attribute if has a Partially Functional Dependency on a PK if its value is determined by **part of the PK**(Composite Key)

- **Transitive Functional Dependency**

Attribute is Transitively Functional Dependency on a table if its value is determined by another **non-key attribute** which it self determined by PK

Example

<u>SID</u>	SName	Birthdate	City	Zip Code	<u>Subject</u>	Grade	Teacher
1	Ahmed	1/1/1980	Cairo	1010	DB	A	Hany
1	Ahmed	1/1/1980	Cairo	1010	Math	B	Eman
1	Ahmed	1/1/1980	Cairo	1010	WinXP	A	khalid
2	Ali	1/1/1983	Alex	1111	DB	B	Hany
2	Ali	1/1/1983	Alex	1111	SWE	B	Heba
3	Mohamed	1/1/1990	Cairo	1010	NC	C	Mona

Full Functional Dependency

Sid,Subject → Grade

Partial Functional Dependency

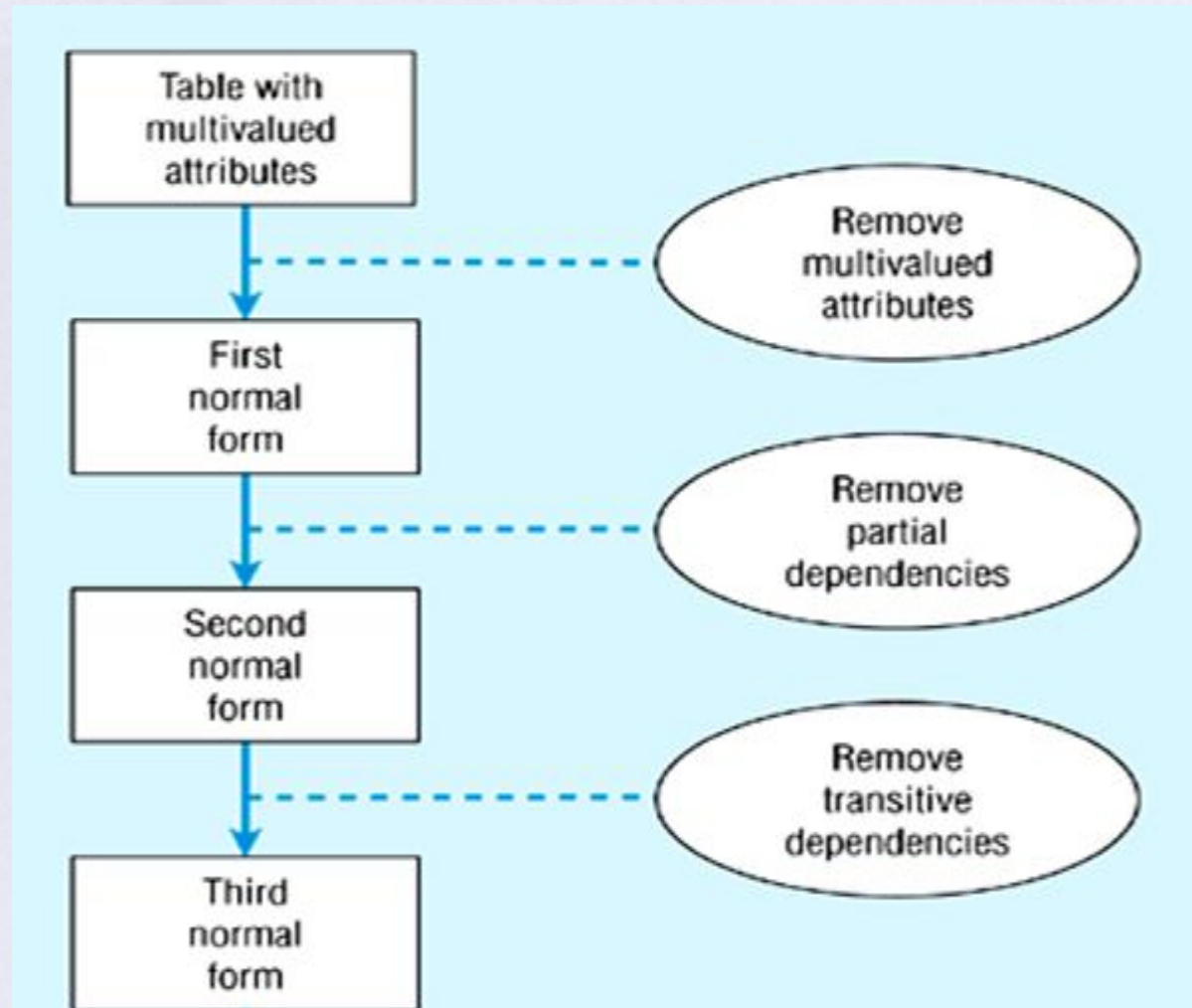
Sid → SName

Subject → Teacher

Transitive Functional Dependency

ZipCode → City

Steps in normalization



1NF

- relation is in **first normal form** if it contains no multivalued or composite attributes
- remove repeating groups to a new table as already demonstrated, “carrying” the PK as a FK
- All columns (fields) must be atomic
 - Means : no repeating items in columns

Example

<u>SID</u>	SName	Birthdate	City	Zip Code	<u>Subject</u>	Grade	Teacher
1	Ahmed	1/1/1980	Cairo	1010	DB	A	Hany
1	Ahmed	1/1/1980	Cairo	1010	Math	B	Eman
1	Ahmed	1/1/1980	Cairo	1010	WinXP	A	khalid
2	Ali	1/1/1983	Alex	1111	DB	B	Hany
2	Ali	1/1/1983	Alex	1111	SWE	B	Heba
3	Mohamed	1/1/1990	Cairo	1010	NC	C	Mona

<u>SID</u>	SName	Birthdate	City	Zip Code	<u>Subject</u>	Grade	Teacher
1	Ahmed	1/1/1980	Cairo	1010	DB	A	Hany
					Math	B	Eman
					WinXP	A	khalid
2	Ali	1/1/1983	Alex	1111	DB	B	Hany
					SWE	B	Heba
3	Mohamed	1/1/1990	Cairo	1010	NC	C	Mona

Repeating Groups
Or multivalued



1NF

Student(SID, SName, Birthdate, City, Zip Code)

<u>SID</u>	SName	Birthdate	City	Zip Code
1	Ahmed	1/1/1980	Cairo	1010
2	Ali	1/1/1983	Alex	1111
3	Mohamed	1/1/1990	Cairo	1010

Stud_Subject (SID, Subject, Grade, Teacher)

<u>SID</u>	<u>Subject</u>	Grade	Teacher
1	DB	A	Hany
1	Math	B	Eman
1	WinXP	A	khalid
2	DB	B	Hany
2	SWE	B	Heba
3	NC	C	Mona

2NF

- a relation is in **second normal form** if it is in first normal form AND every nonkey attribute is fully functionally dependant on the primary key
- i.e. remove partial functional dependencies, so no nonkey attribute depends on just part of the key

2NF

Student(SID, SName, Birthdate, City, Zip Code)

<u>SID</u>	SName	Birthdate	City	Zip Code
1	Ahmed	1/1/1980	Cairo	1010
2	Ali	1/1/1983	Alex	1111
3	Mohamed	1/1/1990	Cairo	1010

2NF

Because there is no
Composite PK

Stud_Subject (SID, Subject, Grade, Teacher)

<u>SID</u>	<u>Subject</u>	Grade	Teacher
1	DB	A	Hany
1	Math	B	Eman
1	WinXP	A	khalid
2	DB	B	Hany
2	SWE	B	Heba
3	NC	C	Mona

SID, Subject → Grade.....FFD

Subject → Teacher.....PFD

2NF

Student(SID, Sname, Birthdate, City, Zip Code)

<u>SID</u>	SName	Birthdate	City	Zip Code
1	Ahmed	1/1/1980	Cairo	1010
2	Ali	1/1/1983	Alex	1111
3	Mohamed	1/1/1990	Mansoura	1210

Stud_Subject (SID, Subject, Grade)

<u>SID</u>	<u>Subject</u>	Grade
1	DB	A
1	Math	B
1	WinXP	A
2	DB	B
2	SWE	B
3	NC	C

Subject (Subject, Teacher)

<u>Subject</u>	Teacher
DB	Hany
Math	Eman
WinXP	khalid
SWE	Heba
NC	Mona

Third Normal Form

- 2NF PLUS *no transitive dependencies*
(one attribute functionally determines a second, which functionally determines a third)

2NF

Student(SID, SName, Birthdate, City, Zip Code)

<u>SID</u>	SName	Birthdate	City	Zip Code
1	Ahmed	1/1/1980	Cairo	1010
2	Ali	1/1/1983	Alex	1111
3	Mohamed	1/1/1990	Cairo	1010

Zip Code → CityTFD

Stud_Subject (SID, Subject, Grade)

<u>SID</u>	<u>Subject</u>	Grade
1	DB	A
1	Math	B
1	WinXP	A
2	DB	B
2	SWE	B
3	NC	C

Subject (Subject, Teacher)

<u>Subject</u>	Teacher
DB	Hany
Math	Eman
WinXP	khalid
SWE	Heba
NC	Mona

3NF

Because there is no Transitive Functional Dependency

3NF

Student(SID, SName, Birthdate,)

<u>SID</u>	SName	Birthdate	ZipCode
1	Ahmed	1/1/1980	1010
2	Ali	1/1/1983	1111
3	Mohamed	1/1/1990	1010

Stud_City(City, Zip Code)

City	<u>Zip Code</u>
Cairo	1010
Alex	1111

Stud_Subject (SID, Subject, Grade)

<u>SID</u>	<u>Subject</u>	Grade
1	DB	A
1	Math	B
1	WinXP	A
2	DB	B
2	SWE	B
3	NC	C

Subject (Subject, Teacher)

<u>Subject</u>	Teacher
DB1	Hany
Math	Eman
WinXP	khalid
DB2	Hany
SWE	Heba
NC	Mona

ITI Example

ITI Students Sheet

Platform Name : SWE

Platform Description: Software Engineering

Graduate Manager: Dr.Baha

Appno	Name	F-code	Faculty	Address	Telno	Grade	Att. Hrs	Sdate
123	Ahmed	SC-phy	Science	Haram	3386842	A	600	14 Sep
124	Mona	Eng-cs	Engineering	Dokki	3389745, 3389744, 5123445	B	591	15 Sep
127	Ali	Com-ac	Commerce	Nasr City	2241593, 2222345	A	550	21 Sep
223	Karim	Med-bio	Medicine	Sheraton	2286845	C	600	14 Sep

1NF :

- **Platform** : pfname , pfdesc , pfManager
- **Students_pf**: pfname, appno, name , faculty , F-Code, address, grade, attd , start_date
- **Std_Tel**: appno, telno

2NF

- **Students:** appno, name , faculty , FCode, address
- **Students_pf:** pfname,appno, grade, attd , start_date

Unchanged Tables

- **Platform :** pfname , pfdesc , pfManager
- **Std_Tel:** appno, telno

3NF

- **Students:** appno, name , FCode, address
- **Fac_majors:** faculty , FCode

Unchanged Tables

- **Platform :** pfname , pfdesc , pfManager
- **Std_Tel:** appno, telno
- **Students_pf:** pfname, appno, grade, attd , start_date

Thank You !!!